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Who Pays for Tests that Should Be Free? A Cross-Sectional Household Survey in Upper Assam

Monmoyuri Dutta
Seema Sharma

Abstract

This study examines the level of awareness of free diagnostic services across tea garden, rural non-tea garden and urban households in Assam and analyses the association between awareness and out-of-pocket diagnostic expenditure and pattern of public versus private diagnostic facility utilisation in different population groups. Based on a purposive sample of households, the study concludes that awareness of free diagnostic services in Assam is critically and equitably low, and The study found critically low awareness of free diagnostic services across all three locality groups. Awareness was positively associated with out-of-pocket diagnostic expenditure, while more aware households showed greater private-facility utilisation. These findings suggest that awareness alone may not translate into financial protection when households perceive limitations in public diagnostic service availability or quality.

Introduction

Accurate and affordable diagnostic services are central to effective healthcare and are widely recognised as a key factor of universal health coverage (UHC). The diagnostic gap in health systems is an increasingly acknowledged concern particularly in resource constrained low- and middle-income regions where poor investment in diagnostic infrastructure has translated directly into high out-of-pocket expenditure for households. Quality diagnostics when integrated into health-care delivery influence disease detection, treatment outcomes and disease surveillance. Henceforth, their absence generates downstream public health consequences including irrational antibiotic use and antimicrobial resistance.

In the year 2018, World Health Organization (WHO) published the first Essential Diagnostics List (EDL) which identifies 113 essential diagnostics that must be done at the primary care level and laboratory-based facilities to place the diagnostics alongside essential medicines as a positioning diagnostic alongside essential drugs as a key component of health system strengthening (WHO, 2018). This global policy signal emphasised the value of country-level framework in assessing and improving diagnostic availability and accessibility.

Under-investment in diagnostics infrastructure has remained a persistent challenge in the public health system in India. Data from the 71st round of the National Sample Survey (NSSO) indicate that diagnostic costs accounted for approximately 10 per cent of all out-of-pocket health expenditure – 11 per cent in outpatient and 9.6 per cent in inpatient settings (Government of India, 2015). In response, India introduced the Free Diagnostics Service Initiative (FDSI) in 2015, the Ayushman Bharat Comprehensive Primary Health Care programme in 2018 and the National Essential Diagnostics List (NEDL) in 2019. While these initiatives mark important progress. Studies consistently report considerable infrastructure gaps, reagent shortages and disparities in implementation across public healthcare settings, particularly in relation to non-communicable diseases and point-of-care diagnostics.

The persistence of diagnostic OOPE may reflect not only the absence of services but a more complex structural failure. Households cannot effectively navigate a fragmented diagnostic ecosystem characterised by multiple providers, referral barriers, administrative complexity and quality concerns. Awareness of free diagnostic entitlements should, therefore, be understood not merely as knowledge of service availability but as the capacity to translate that knowledge into utilisation. The health system failure to communicate and operationalise free diagnostic entitlements meaningfully shifts the financial burden back on households, despite services being officially available and available free of cost.

Despite growing emphasis on strengthening diagnostic infrastructure and broadening policy coverage, relatively little is understood about the ways households perceive, understand and respond to these entitlements which is a question of particular importance in geographically and socioeconomically marginalised regions such as Upper Assam, where tea garden communities face distinct structural barriers to healthcare communication and access.

This study is conceptually situated within Behavioural Model of Health Services Use of Andersen which conceptualises healthcare utilisation as a function of predisposing, enabling, and need factors acting in combination (Andersen, 1995). Within this framework, awareness of free diagnostic entitlements functions as a predisposing factor while financial resources, health insurance coverage and locality-specific infrastructure and referral conditions operate as enabling factors that influence the extent to which such awareness is converted into actual service use. Perceived clinical need for diagnostic testing constitutes the third determinant interacting with predisposing and enabling factors that influence shape eventual facility choice and OOPE. Framing the present inquiry in these terms enables the relationship between awareness and OOPE to be interpreted beyond a simple knowledge deficit but as the outcome of interacting predisposing, enabling and need-based determinants of health-seeking behaviour.

Research Gap

Previous research in this area has focused predominantly on policy design and supply-side infrastructure, leaving the household-level relationship between the

awareness about free diagnostic entitlements and out-of-pocket diagnostic expenditure empirically under examined within the Indian public health landscape. The existing literature has yet to explore this relationship comparatively across tea garden, rural non-tea garden and urban populations in a single sampling frame. This is especially pressing in Upper Assam, where tea garden communities face distinct barriers to healthcare communication and access.

Aims and Objectives

To address the identified gaps, this study focuses on three specific research questions. First, it examines the extent of awareness of free diagnostic services among surveyed households and how does the awareness vary across tea garden, rural non-tea garden and Urban households of Upper Assam. Second, it examines the association of the awareness about diagnostic services with out-of-pocket diagnostic expenditure. Finally, the study explores whether households aware of free diagnostic schemes preferentially use public diagnostic facilities over private ones or not.

Literature Review

India introduced the Free Diagnostics Service Initiative (FDSI) in 2015 to improve access to essential diagnostic services and reduce OOPE. The initiative was further expanded under Ayushman Bharat in 2018. However, despite these policy measures, implementation gaps and infrastructural limitations continue to constrain equitable diagnostic access (Padhan and Padhy, 2025).

Evidence from a pilot facility survey across Karnataka, Uttar Pradesh and Maharashtra documented considerable inter-district variation in availability of diagnostic services with blood count test, glycated haemoglobin test, HIV and syphilis diagnostic and tuberculosis and malaria microscopy unavailable in many primary health centres (Kruk et al, 2018). The study noted that public-private partnership models were linked to relatively better diagnostic availability.

A comprehensive synthesis of Common Review Mission reports under National Health Mission (NHM) from 2007 to 2021 has revealed that despite improved service availability under the NHM, household OOPE on diagnostics remains high. Mean OOPE was estimated at $74.7 for inpatient and $37.9 for outpatient services at private facilities compared to $11.8 and $22.4 respectively at public hospitals (Hannah et al, 2023). This study illustrates the disproportionate cost burden of private diagnostic care.

It has been observed in a study that the share of OOPE attributable to diagnostic tests increased with the socioeconomic status (Ambade et al, 2022). The National Health Accounts data, however, indicate that OOPE as a proportion of overall health expenditure declined from 64.2 per cent in 2013-14 to 39.4 per cent in 2021-22 (Government of India, 2024), but disaggregated expenditure in diagnostics in marginalised communities including tea garden population remains sparse.

There is a broad agreement across different studies that diagnostic OOPE in India has remained substantial despite more than a decade of policy intervention and that availability gaps and quality concerns continue to push households toward costlier private alternatives (Kruk et al, 2018; Hannah et al, 2023; Ambade et al, 2022). A notable concern, nonetheless, emerges between declining national OOPE share reported in aggregate health accounts and persistently high and even regressive, diagnostic-specific burden documented at the facility and household level. This suggests that macro-level improvement may mask micro-level inequities (Government of India, 2024; Ambade et al, 2022). However, none of these studies have disaggregated diagnostic OOPE by population subgroups leaving open whether marginalised communities such as tea garden population experience this burden differently and whether its persistence reflects availability constraints or failure in awareness and entitlement utilisation or both.

Compounding these supply-side constraints, Padhan and Padhy (2025) identify inadequate public awareness as a persistent barrier to the equitable utilisation of free diagnostic services irrespective of the extent to which services are formally provisioned. Similarly, studies related to Rashtriya Swasthya Bima Yojana (RSBY) have consistently documented that high enrolment rates do not translate into utilisation because households, particularly in rural and remote settings, lack actionable knowledge of their entitlements, eligibility criteria and access pathways (Selvaraj and Karan, 2012). The findings additionally suggest that insurance schemes in Tamil Nadu may have paradoxically increased OOPE for the poor given their exclusion of outpatient diagnostic and drug costs.

These two strands of evidence agree on a common mechanism – nominal awareness or scheme enrolment without actionable knowledge of entitlements and access pathways, fails to produce utilisation or financial-protection gains (Padhan and Padhy, 2025; Selvaraj and Karan, 2012). They also highlight that enrolment can paradoxically increase OOPE where coverage is incomplete, challenging any simple equation of scheme presence with protection. What remains unexamined is whether this awareness and utilisation of OOPE disjuncture well documented for health insurance schemes nationally replicates specifically for free diagnostic entitlements and whether it holds within structurally distinct, marginalised settings such as tea garden communities of Assam.

In the context of Assam, an investigative analysis has documented that tea garden workers have to travel to nearby towns for healthcare under compounding barriers of remote location, inadequate roads and poor socioeconomic conditions. Women reported being charged for diagnostic tests that should be free at government hospitals. (IndiaSpend, 2024).

The investigative and institutional evidence from Assam agrees with and plausibly intensifies the national-level pattern established above. Structural isolation, absence of clinical staff and weak government communication compound the supply-side and awareness-related failures while household-level testimony confirms that nominally free entitlements are routinely not honoured at the point of care. This evidence base, however, remains qualitative and institutional rather than systematic and quantitative. To the best of our knowledge, there is no study that has statistically compared awareness of free diagnostic services, diagnostic OOPE and facility-utilisation patterns across tea

garden, rural non-tea Garden and urban households within a single, comparable, sampling frame.

Considering together, the literature establishes that the persistence of diagnostic OOPE in India reflects an interaction between supply-side failures and demand-side information gap which is plausibly intensified within the tea garden ecosystem of Assam by structural isolation.

Methods

Research Methodology

The study is based on a cross-sectional household survey carried out in a purposely selected ample of 400 households from rural tea garden, rural non-tea garden and urban areas in six districts of Upper Assam India – Tinsukia, Dibrugarh, Sivasagar, Charaideo, Jorhat and Golaghat. These districts account for approximately 70 per cent of 879 tea gardens in Assam and represent the core plantation belt of the state. The selection of the districts and households with districts was purposive based on their demographic, economic and institutional significance within the tea sector and high concentration of tea garden healthcare infrastructure in these districts including facilities under the public-private partnership (PPP) model. Only those households were included in the sample which had incurred at least one healthcare expenditure in the preceding three months of the survey. Half of the households covered under the survey were rural tea-garden households while the remaining half of the households were divided equally between rural non-tea garden and urban households. In each household, data were collected preferably from the head of the household head. If the household head was not available data were collected from the most informed adult member of the household who also provided the self-rated awareness assessment described below. Because these variables were captured through one respondent per household rather than aggregated across all household members, the terms “household” and “respondent” are used interchangeably throughout this paper to refer to the same sampled unit. Reported awareness levels and out-of-pocket expenditure accordingly are, therefore, based on the knowledge, recall and reporting by the respondent. This proxy-reporting design is revisited as a limitation below.

A structured, pre-tested questionnaire comprising of two modules was administered to collect the data required for the study. The first module was related to the household sociodemographic profile while the second module was related to healthcare expenditure during three months prior to the survey to minimise recall bias while capturing sufficient variation in healthcare utilisation.

Awareness about free diagnostic services was captured through two components of the questionnaire. A single global item asked respondents to self-rate their awareness about government health schemes in Assam on a four-point categorical scale: “Fully aware,” “Heard but don’t know details,” “Not aware” and “Not sure.” This item generated the primary awareness classification. A supplementary multi-item checklist recorded name-recognition across 14 named state and central schemes, including Chief Minister

Free Diagnostic Services scheme most directly relevant to this study. Eight items from this checklist constituted the "Health Scheme Awareness" scale, which was subjected to internal-consistency testing during instrument development (Cronbach's $\alpha = 0.81$; "Good," per George and Mallery, 2003), as part of a broader psychometric screening of 35-items instrument. "Full awareness" as used in this paper, therefore, denotes global self-rating by the respondents rather than a composite score of verified knowledge. The instrument did not separately probe awareness of eligibility criteria, of which specific tests are free or where free diagnostic services can be accessed.

Primary data were collected through direct interview conducted out by the authors. Pilot testing was undertaken on a small convenience sample to assess questionnaire clarity and estimate interview duration with appropriate modifications implemented before full-scale deployment. Informed verbal and written consent was obtained from each participating household. All completed questionnaires were reviewed daily for completeness and consistency.

All data were compiled, cleaned and analysed using IBM SPSS. Statistical methods used for analysis included Pearson χ^2 tests for categorical association, Kruskal-Wallis H tests for non-normally distributed ordinal and continuous variables, Spearman rank-order correlation for the association between awareness and OOPE and binary logistic regression to identify independent predictors of public versus private facility use. Non-normality of expenditure distributions was confirmed using Shapiro-Wilk and Kolmogorov-Smirnov tests. Descriptive statistics included means, standard deviations, medians and maximum values.

Results

Awareness about Free Diagnostic Services

Sensitization about government-provided free diagnostic services was critically low across all the three subgroups as Only 1.0 per cent reported full awareness of available diagnostic schemes, while 52.3 per cent had heard of such schemes but lacked specific knowledge of entitlements, eligibility criteria or access pathways. Another 25.3 per cent reported no awareness, while 19.5 per cent remained uncertain. These findings reflect a pattern indicating a profound information deficit that undermines scheme effectiveness irrespective of service supply.

Tea garden households exhibited the highest proportion reporting no awareness (33.5%), compared with 19.5% among rural non-tea garden and 13.8% among urban households. Urban households recorded the highest proportion reporting full awareness (3.4%), although this remained small in absolute terms. The 'not sure' category was most prevalent among rural non-tea garden households (25.7%), possibly reflecting partial exposure to information without actionable knowledge. The Pearson χ^2 test confirmed that the categorical distribution of awareness differed significantly across locality subgroups ($\chi^2(8) = 27.295$, $p < 0.001$). However, the Kruskal-Wallis H test of ordinal awareness level was statistically insignificant ($H=0.428$, $df=2$, $p=0.808$) with near-

identical mean ranks across rural tea garden (206.28), rural non-tea garden (207.80) and Urban (198.47) subgroups and a common median score of 2 (Heard but do not know details). The findings show that, while the categorical composition of awareness differs across subgroups, the overall depth of awareness is equally poor everywhere.

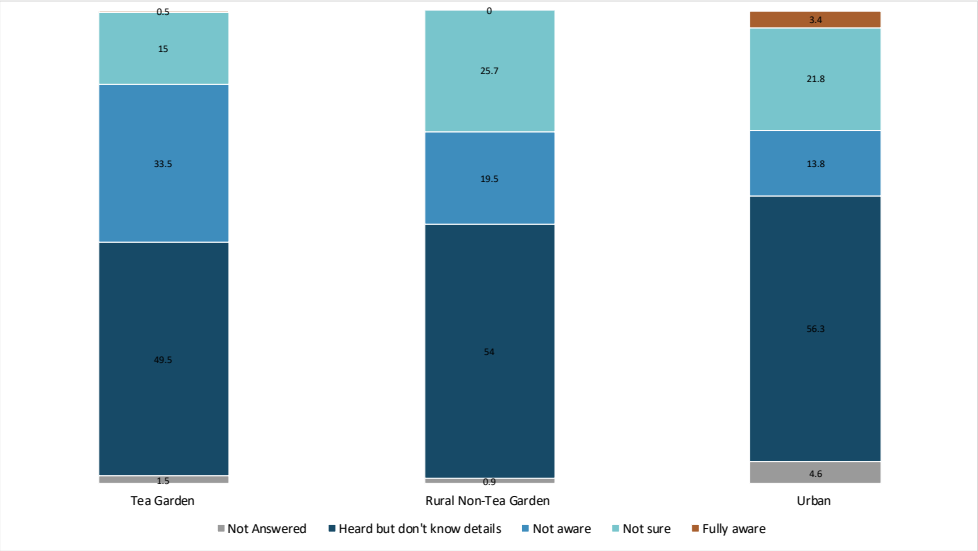


Figure 1: Level of awareness about free diagnostic services across different subgroups of respondents.
Source: Authors

Table 1: Level of awareness about free diagnostic services across different subgroups of respondents.

Awareness Level	Tea Garden	Rural non-Tea Garden	Urban	Total
Not Answered	1.5%	0.9%	4.6%	2.0%
Heard but don't know details	49.5%	54.0%	56.3%	52.3%
Not aware	33.5%	19.5%	13.8%	25.3%
Not sure	15.0%	25.7%	21.8%	19.5%
Fully aware	0.5%	0.0%	3.4%	1.0%
N	200	113	87	400

Source: Authors

Association Between Awareness and OOP Diagnostic Expenditure

Counter to the expectation that awareness about free diagnostic services would reduce out-of-pocket diagnostic expenditure, the average OOP diagnostic expenditure is found to be the highest in the fully aware subgroup, substantially exceeding the 'heard but do not know details' group and 'not aware' group in which the average expenditure was found to be the lowest across all awareness groups.

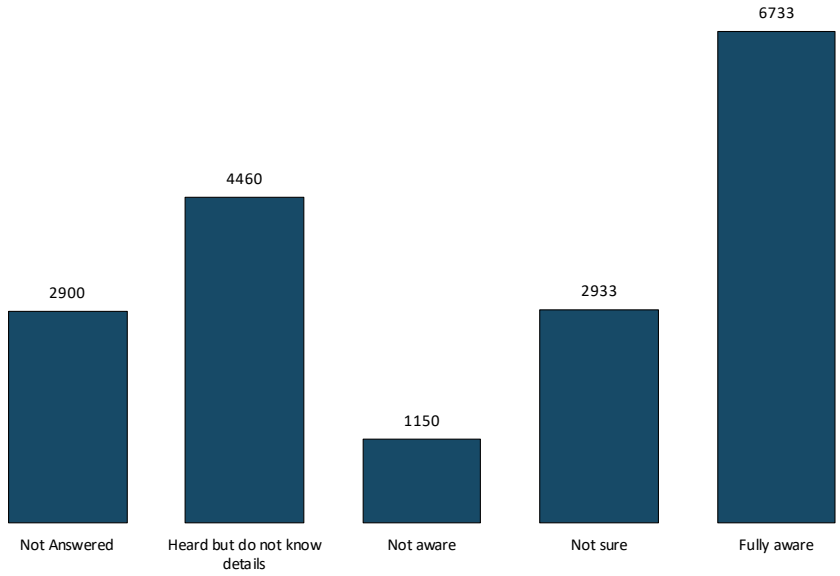


Figure 2: OOP diagnostic expenditure (Rupees) by the level of awareness.
Source: Authors

Table 2: OOP diagnostic expenditure (Rupees) by the level of awareness about free diagnostic schemes.

Awareness level	N	Number used private facility	OOP expenditure (Rupees)		
			Average	Minimum	Maximum
Not Answered	8	2	2900	2900	2900
Heard but do not know details	209	23	4460	1300	12000
Not aware	101	4	1150	300	2700
Not sure	78	17	2933	700	6000
Fully aware	4	3	6733	2300	12000
Total	400	49	3736	300	12000

Source: Authors

Diagnostic expenditure arises only when a paid service is purchased. The comparison across awareness groups was therefore restricted to those 49 respondents who paid for private diagnostic services. The rest either used no diagnostic services or used public facilities without paying and counting them as zero-expenditure cases would have treated non-utilisation as financial protection.

Because the expenditure distribution was non-normal (Shapiro-Wilk and Kolmogorov-Smirnov tests), the Kruskal-Wallis H test was applied to this restricted sample. It found no statistically significant difference in OOP diagnostic expenditure

across awareness groups (H=4.993, df=4, p=0.288). The pattern of mean ranks, however, runs in the same direction as in the full sample: the highest among fully aware respondents (42.17) and the lowest among those who had heard of the schemes without knowing the details (22.78) with the unaware (27.13) and unsure (24.53) groups in between. The three middle categories are closely clustered and the high rank of the fully aware group rests on only three respondents. Among households that used private diagnostic facilities on payment, it is not possible to ascertain whether awareness about free diagnostic schemes resulted in lowering OOP diagnostic expenditure because of the very small sample.

Table 3: Kruskal–Wallis rank of diagnostic expenditure among private-facility users by awareness level.

Awareness level	N	Mean Rank
Not Answered	2	24.50
Heard but don't know details	23	22.78
Not aware	4	27.13
Not sure	17	24.53
Fully aware	3	42.17

Source: Authors

Public and Private Diagnostic Facility Utilisation

The overall distribution of the facility use was heavily weighted toward public facilities (87.8 per cent), with 12.3 per cent of the respondents seeking care in private facilities during their latest healthcare episode. However, this aggregate pattern conceals a striking reversal among fully aware households – 75.0 per cent of these respondents used private facilities compared with 25.0 per cent of those who used public facilities. Private facility utilisation was the least common among respondents who were not sure about the schemes and those who were unaware about the schemes. The Pearson χ^2 test confirmed the association between awareness level with public and private facilities utilisation ($\chi^2 = 29.231$, df=4, $p<0.001$), though very few respondents who were fully aware limit categorical inference.

Table 4: Use of public and private facilities by the level of awareness among responents.

Awareness level	Private facility	Public facility	N
Not Answered	25.00	75.00	8
Heard but do not know details	11.00	89.00	209
Not aware	4.00	96.00	101
Not sure	21.80	78.20	78
Fully aware	75.00	25.00	4
Overall	12.30	87.80	400

Source: Authors

To test the relationship between the awareness about free diagnostic schemes and use of private or public facilities, we initially carried out the conventional binary logistic regression analysis. The dependent variable for the regression analysis was a categorical variable which has value 0 if the respondent used a private facility and value 1 if the respondent used a public facility. Covariates in the analysis included age of the re-

spondent, household size and household asset index while categories included awareness about free diagnostic schemes, level of education of the household head, any member of the household covered by health insurance, gender, whether household had a BPL card and locality. Table 5 shows the codes assigned to different categories of the categorical variables along with the specification of the reference category. The model, however, exhibited complete separation, particularly because there was no private-facility user among household heads who were uneducated. A similar separation problem was observed as there was no private facility use in households with a BPL card. The conventional logistic regression therefore failed to converge and produced unstable estimates. To address this estimation problem appropriately, we have used Firth penalised logistic regression (Firth, 1993). Firth penalised logistic regression is a bias-reduction method that fixes standard maximum likelihood estimation failures when the sample size is small and provides finite bias-reduced estimates in the presence of complete or quasi-complete separation. This approach is specifically recommended for situations in which standard maximum-likelihood logistic regression cannot obtain reliable finite parameter estimates (Suhas et al, 2023).

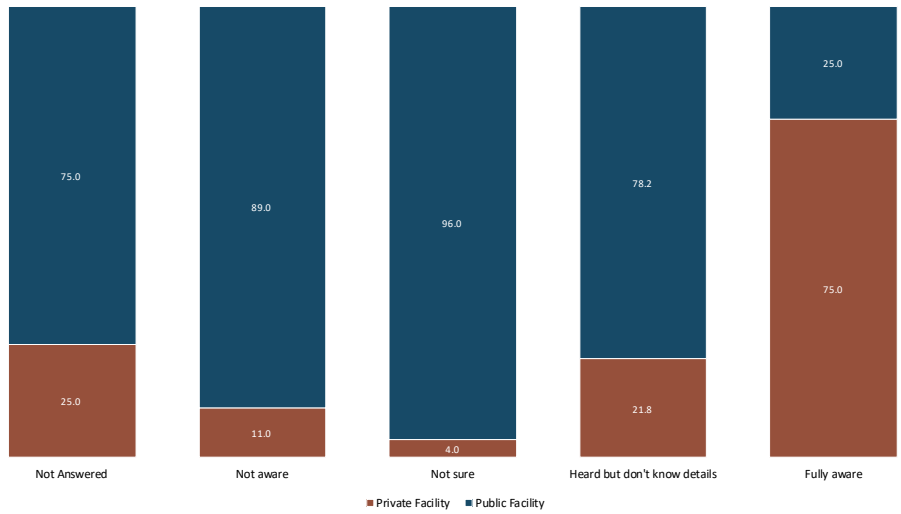


Figure 3: Use of public and private facilities by the level of awareness.
Source: Authors

Results of the regression analysis are presented in table 6. The analysis confirms that awareness about free diagnostic schemes is a statistically significant predictor of the use of public facilities. Households having no awareness about free diagnostic schemes are 66 per cent less likely to use public facilities instead of private facilities when compared with households having any awareness about free diagnostic schemes. The other predictor that has been found to be statistically significant is whether any member of the household had a health insurance. Households in which no member of the household had a health insurance were more than nine times more likely to opt for a public facility instead of a private facility when compared to households in which at least

one member had a health insurance. The household asset index has also been found to be associated with the use of public facilities – the higher the household asset index the lower the probability of using a public facility. Rest of the explanatory variables have not been found to be statistically significantly associated with the use of a public facility relative to the use of a private facility.

Table 5: Specification of categorical explanatory variables used in the analysis.

Variable	Category	Code	
Awareness about free diagnostic schemes	No awareness	1	
	Any awareness	2	Reference category
Education of household head	Uneducated	1	
	Educated up to high school	2	
	Educated above high school	3	Reference category
Gender	Male	1	
	Female	2	Reference category
Household having BPL card	No	1	
	Yes	2	Reference category
Locality	Rural tea garden	1	
	Rural	2	
	Urban	3	

Source: Authors

Discussion

This cross-sectional study has investigated the three interconnected facets of the utilisation of free diagnostic services in 400 households selected purposively from different regions of Upper Assam: 1) level of household awareness about free diagnostic services provided by the government; 2) association between the awareness about free diagnostic services and out of pocket (OOP) expenditure on diagnostic services; and 3) pattern of public versus private diagnostic facility utilisation by the level of awareness about free diagnostic services of the government. The findings of the study, collectively, reveal the existence of a structural paradox. The government scheme of free diagnostic services which is designed to eliminate diagnostic costs are largely unknown to their intended beneficiaries. There is only a small proportion of the households surveyed that are aware about this scheme and these households are incurring disproportionately high expenditure on diagnostic services by preferring private over public diagnostic facilities. This structural paradox has important implications for health equity and the design of demand-side health financing interventions in Assam that needs to be addressed at the policy level. There is little evidence to suggest that availability of free diagnostic services in public institutions has contributed to reducing the cost of diagnostics.

Critically Low and Inequitably Distributed Awareness

The most immediate finding is a near-universal absence of functional awareness about free diagnostic schemes across the households surveyed. Only 1.0 per cent of the respondents reported full awareness of government-sponsored free diagnostic services,

while 52.3 per cent had merely heard about these schemes but had little knowledge of entitlements, eligibility criteria or access pathways. A further 25.3 per cent reported no awareness and 19.5 per cent remained uncertain. This distinction between nominal and actionable awareness is analytically critical (Selvaraj et al, 2019). It aligns with broader evidence suggesting that demand-side information, education, communication (IEC) strategies in India generate nominal awareness so that they fail to transfer the knowledge that is necessary to change behaviour (Baru et al, 2010; Reddy et al, 2021). The proportion reporting no awareness was highest among tea garden households (33.5 per cent), compared with 19.5 per cent among rural non-tea garden and 13.8 per cent among urban households. This awareness gap may be attributed to the structural insularity of the plantation ecosystem, although this study has not directly measured such structural factors, but it limits the fidelity of the government health communication (Gupta et al., 2016; Bhattacharya, 2018). The deficiency in effectively informing and communicating with the people is also reflected from the finding that the common median of all the three localities – tea garden, rural non-tea garden, urban – was 'heard but do not know details'. This convergence indicates a systemic information deficit among the households surveyed rather than a geographically localised one.

Table 6: Determinants of the use of free public diagnostic facilities relative to the private diagnostic facilities.

Predictor	Reference category	Adjusted OR	95 CI	p
Age	Continuous	1.024	0.981–1.069	0.272
Household size	Continuous	1.061	0.676–1.665	0.796
Asset index	Continuous	0.755	0.615–0.927	0.007
Awareness of free diagnostic schemes	No awareness	0.339	0.149–0.772	0.010
	Any awareness®			
Education of household head	Uneducated	0.656	0.033–12.865	0.781
	Up to High school	0.273	0.014–5.449	0.396
	Beyond High school®			
Gender	Male	1.273	0.429–3.781	0.664
	Female®			
BPL Card	No	8.569	0.637–115.241	0.105
	Yes®			
Any member covered by health insurance	No	9.458	2.473–36.177	0.001
	Yes®			
Locality	Rural Non–Tea Garden	1.179	0.208–6.681	0.852
	Tea Garden	0.637	0.267–1.516	0.308
	Urban®			

® Reference category
Source: Authors

The Awareness-Expenditure Paradox

Contrary to the theoretical premise of demand-side health promotion which argues that awareness about free services leads to increased utilisation of these services and thereby reduces OOP expenditure, the present analysis reveals a significant positive association between the level of awareness of diagnostic schemes and the OOP expenditure. The OOP expenditure has been found to be the highest in those households where the awareness about the free diagnostic schemes was the highest. Two intersecting mechanisms may plausibly underlie this paradox, although the cross-sectional design of the study does not allow to directly test this hypothesis. Instead, both observations should be read as possible explanations rather than demonstrated causes. First, a possible endogeneity dynamic, viz awareness about free diagnostic schemes is not randomly distributed across the sample but is concentrated among households with greater health-seeking propensity and more extensive engagement with the formal health system, which would make it a marker of healthcare engagement rather than an independent predictor of lower expenditure (Nandi et al, 2017). Second, a possible quality exit dynamic whereby even households with full knowledge of their entitlements, choose for private diagnostic facilities when public services are perceived as inadequate in test availability, reagent supply and result timeliness (Prinja et al, 2019; Garg et al, 2020). The insignificance of the health insurance as a moderating factor in this relationship reinforces the observation that the insurance architecture as currently designed is insufficient to redirect households aware about free diagnostic services towards free public diagnostics services (Selvaraj and Karan, 2012; Ghosh, 2014).

Facility Utilisation: The Informed Exit

The analysis of the public and private diagnostic facility use by the level of awareness about free diagnostic scheme provides the pattern most consistent with a possible quality exit dynamic though perceived service quality was not directly measured in this study. While 87.8 per cent of households used public diagnostic facilities, 75.0 per cent of the fully aware households (3 out of 4) reported using private diagnostic facilities. This inversion was found to be statistically significant ($\chi^2=29.231$, $df=4$, $p<0.001$), although the very few households fully aware of free diagnostic scheme limits the stability of this categorical comparison. This inversion mirrors the informed exit documented in analogous settings where more health-literate populations preferentially seek private care upon developing a critical appraisal of public service quality (Mohanty & Srivastava, 2013; Rao et al, 2011).

Firth penalised logistic regression analysis has indicated that structural and socioeconomic factors are associated with the utilisation of public and private diagnostic facilities. The higher level of the corrected asset index was found to be associated with lower odds of the use of public diagnostic facilities. Similarly, any type of awareness about free diagnostic scheme was found to be associated with higher odds of the use of public diagnostic facilities compared to households having no awareness about the scheme. Health insurance coverage was strongly associated with the use of private diagnostic facilities. Education, gender, BPL status and locality were not statistically significant after adjustment.

The lower odds of public-facility use among households with awareness are consistent with the descriptive pattern in which more aware households showed greater private-facility use. However, the cross-sectional design does not establish why aware households were more likely to use private facilities, and perceived service quality was not directly measured.

Limitations of the Study

Several limitations qualify the findings of this study. First, the cross-sectional design precludes causal inference. The positive awareness-expenditure association may reflect reverse causation whereby households with greater illness burden simultaneously accumulate scheme awareness through repeated healthcare contact and higher OOP diagnostic expenditure rather than any directional effect of awareness on OOP. Second, the awareness variable measures knowledge of government health schemes broadly rather than free diagnostic programmes specifically risking misclassification of respondents familiar with unrelated entitlements. The study did not assess whether respondents knew their eligibility for free diagnostics, which specific tests are covered or where such services can be accessed. The "full awareness" category should, therefore, be read as an upper-bound rather than a validated measure of functional or actionable awareness. Third, self-reported OOP expenditure over a three-month recall period is susceptible to telescoping and recall bias that may introduce non-differential measurement error that attenuates true associations. Fourth, the fully aware subgroup is very small (n=4), which limits the statistical stability of findings. Finally, the study covers only three locality types, and the findings may not be generalised to tribal belt areas, riverine char communities or states operating under different scheme architectures. The purposive sampling also limits the population-representativeness of the findings. Sixth, because a single adult household member reported on behalf of all members of the household, the reported findings reflect the respondent knowledge and recall rather than a verified household-wide consensus or within-household variation in awareness or in expenditure decision-making.

Policy Implications

The findings carry several implications for health policy in Assam. First, the distinction between nominal and actionable awareness underscores that IEC strategies must move beyond name recognition to provide actionable, step-by-step guidance on eligibility, access pathways and entitlements under the Free Diagnostic Services Initiative (FDSI) of the government of Assam. Second, if the possible quality exit dynamic suggested in the present study holds, it would imply that demand-side awareness interventions cannot succeed without commensurate supply-side investment. NQAS accreditation of public laboratories, reagent supply chain strengthening, expanded the list of test and timely result reporting are prerequisites for retaining even well-informed patients within the public system. Third, tea garden communities require estate-specific, targeted interventions such as mobile diagnostic camps within estates, management-level engagement and trilingual IEC materials that navigate the communication barriers of the

plantation ecosystem. Fourth, the dominant role of health insurance in predicting public facility use reinforces that structural financing mechanisms exert greater influence than information alone in shaping utilisation pathways. The diagnostic benefit packages under AB-PMJAY and state schemes should be reviewed to reduce administrative barriers and extend empanelment to government diagnostic centres. Fifth, sub-district equity monitoring that disaggregates tea garden and rural non-tea garden population should be institutionalised within HMIS and NHM district planning frameworks to enable targeted policy responses.

Conclusion

This study offers important insights into a critical disconnect at the heart of the diagnostic ecosystem of Assam. It reveals that mere existence of free diagnostic schemes is insufficient to reduce out-of-pocket expenditure on diagnosis when awareness remains shallow, supply-side quality is possibly inadequate and access pathways are procedurally complex. Awareness about free diagnostic services was critically low across the surveyed households and differed significantly in categorical composition across locality subgroups. Contrary to the expected protective effect, greater awareness was associated with higher diagnostic out-of-pocket expenditure and lower odds of public-facility utilisation. The findings suggest that awareness alone may be insufficient to generate financial protection when public diagnostic services are perceived as inadequate or difficult to access.

The three findings of the present analysis converge on a single policy conclusion – the demand-side awareness efforts and supply-side quality investment must be advanced in concert. The realisation of universal diagnostic access in Assam requires a coordinated policy response in which financing reform, quality improvement and equity-sensitive communication together enable awareness generation. While this study is limited by its cross-sectional design and purposive sample design, small size of fully aware subgroup and absence of illness severity data, it establishes a clear evidence base for the urgent redesign of both IEC strategies and public diagnostic service delivery in Upper Assam.

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